OLIVIER CÔTÉ, ACAS

PhD Candidate in Actuarial Science, Université Laval

CONTACT INFORMATION

Québec, QC, Canada
olivier.cote.12@ulaval.ca
in/olivier-cote-act
olicoside.github.io

SKILLS

R	8 years +
Python	3 years +
LaTeX	6 years +

LANGUAGES

Français	Mother tongue
English	Advanced
Svenska	Beginner, <1 yr

EDUCATION

PhD in Actuarial Science (GPA : 4.17/4.33) — 2023 - 2027
Université Laval
Thesis: *A causal perspective on direct and indirect discrimination based on sensitive characteristics in insurance predictive models*. Supervised by Prof. Marie-Pier Côté (Université Laval) and Prof. Arthur Charpentier (Université du Québec à Montréal), in a Mitacs Accelerate partnership with one of the largest P&C insurers in Canada. Initial thesis deposit planned for March 2027; defence expected in summer 2027.

Master's degree in Actuarial Science, research-based (GPA : 4.24/4.33) — 2021 - 2023
Université Laval
Supervised by Prof. Marie-Pier Côté and Prof. Arthur Charpentier.

Bachelor's degree in Actuarial Science (GPA : 4.17/4.33) — 2018 - 2021
Université Laval

Research visits — 2023 - 2026
Stockholm University, University of Toronto, Bayes Business School

- Department of Mathematics, Stockholm University (two weeks, January 2026): invited seminar and research discussions with Prof. Mathias Millberg Lindholm.
- Department of Statistics, University of Toronto (2025): invited seminar.
- Bayes Business School, London (one week, 2023): research discussions with Prof. Andreas Tsanakas.

RESEARCH PAPERS

Côté, Olivier, Côté, M.-P., and Charpentier, A. (2027). A statistical perspective on fairness in insurance pricing. *Annual Review of Statistics and Its Application*. Invited review, accepted June 2026. In press

Charpentier, A., **Côté, Olivier**, and Côté, M.-P. (2026b). Governing fairness in insurance pricing: The trilemma of actuarial adequacy, solidarity, and causal legitimacy. *Journal of Business Ethics*. Published online September 2026, doi:10.1007/s10551-026-06479-2

Côté, M.-P., **Côté, Olivier**, and Charpentier, A. (2026). Recoverability of market-wide fair insurance premiums under selection bias. *Insurance: Mathematics and Economics*, 129:103268. doi:10.1016/j.insmatheco.2026.103268

Côté, Olivier, Côté, M.-P., and Charpentier, A. (2026). A scalable toolbox for exposing indirect discrimination in insurance rates. *CAS Research Paper Series on Bias and Insurance*. Research paper, casact.org; code released with a simulated worked example

- Winner of the American Academy of Actuaries Award for Research 2026
- Selected by the National Association of Insurance Commissioners (NAIC) for its 2026 webinar series for U.S. insurance regulators and their actuarial staff

Charpentier, A., Fernandes Machado, A., **Côté, Olivier**, and Hu, F. (2026a). Federated measurement of demographic disparities from quantile sketches. *arXiv:2602.18870*. Preprint

Côté, Olivier, Côté, M.-P., and Charpentier, A. (2025). A fair price to pay: Exploiting causal graphs for fairness in insurance. *Journal of Risk and Insurance*, 92(1):33–75. doi:10.1111/jori.12503

- Finalist for the American Academy of Actuaries Award for Research 2025

WORK IN PROGRESS

Côté, Olivier, Charpentier, A., Côté, M.-P., and Il Idrissi, M. A definition of confounding bias for arbitrary statistical functionals. Working paper, in preparation.

Côté, Olivier, Côté, M.-P., Charpentier, A., and Il Idrissi, M. Profiling actuarial discrimination via causal decomposition. Working paper, presented at the 29th International Congress on Insurance: Mathematics and Economics, Seoul (2026).

DIVERSITY, EQUITY, AND INCLUSION EDUCATION

Get Comfortable Being Uncomfortable: Dialogue About Favoritism and Fairness — 2025
Canadian Statistical Sciences Institute

Let's talk about race and collective responsibility — 2024
Canadian Statistical Sciences Institute

Acting as a sentinel for the prevention of suicide — 2023
Centre de prévention du suicide du Québec workshop

Addressing Conflicts Related to Bias, Privilege, and Identity in the STEM Fields — 2022
CANSSI and Academic Impressions workshop

PROFESSIONAL DESIGNATIONS

Associate of the Casualty Actuarial Society (ACAS) — 2025
Casualty Actuarial Society

Exam 6C — 2025
Casualty Actuarial Society

Professionalism Workshop — 2025
Canadian Institute of Actuaries

Online Course 2 — 2024
Casualty Actuarial Society

MAS-I, MAS-II, Exam 5 and Online Course 1 — 2020 - 2022
Casualty Actuarial Society

Exam P, FM and IFM — 2019- 2020
Society of Actuaries and Canadian Institute of Actuaries (credits)

WORK EXPERIENCES

Course lecturer 2022 - 2023
Université Laval

- Linear Models in Actuarial Science (mandatory undergraduate course, 3 credits)
  - Fall 2022, 88 students enrolled
- Statistical Learning in Actuarial Science (mandatory undergraduate course, 3 credits)
  - Winter 2023, 54 students enrolled

*__Awards included:__ Faculty of Science and Engineering Teaching Star, one per course (fall 2022 and winter 2023).

American football coach 2019 - 2025
Falcons of Cégep de Lévis
American football coach for different teams for more than ten years (2012–2025), including seven years in collegiate football at the Cégep de Lévis (2019–2025):

- Tight ends and Fullbacks coach (2019-2025), responsible for the development of 10 athletes
- Supervisor for a bachelor's degree intern in sports intervention (2023)
- Volunteer mathematics tutor for student-athletes

*__Awards included:__ RSEQ Coaching Staff of the Year 2023, 2024; Sportsmanship Award 2022 (D1), 2023 (D2).

Teaching assistant 2019 - 2024
Université Laval
Teaching assistant on several occasions:

- Evaluator for actuarial student internships (1 time)
- Probabilistic Analysis of Actuarial Risks (3 times)
- Statistical Learning in Actuarial Science (5 times)
- Linear Models in Actuarial Science (1 time)

Actuarial intern 2019 - 2021
Desjardins General Insurance Group
Four internships, May 2019 to August 2021: Property Ratemaking (2019), Business analysis (2020 and 2021), and Research (2021).

SUPERVISION AND MENTORING

Mentor of a master's student, University of Texas 2025 - 2026
On the recommendation of a Hickman Scholars Program colleague
Advised the student's essay on an insurance regulatory issue, and connected the student with a Texas insurance regulator.

Co-supervisor of a master's student, Université Laval 2023
With Prof. Marie-Pier Côté
One-year project on fairness in insurance; the student now works as an actuary in the insurance industry.

Supervisor of a PhD student, Université Laval Ongoing
Mitacs Accelerate partnership
Supervised the PhD student who continues the industrial partnership, and helped design the student's thesis project.

ENGAGEMENT WITH PRACTICE AND POLICY

Online course on fair insurance pricing, English and French versions — 2025 - 2026
Casualty Actuarial Society (in post-production)
Commissioned by the CAS; with Marie-Pier Côté and Arthur Charpentier. We defined the learning objectives and recorded 50 minutes of two-minute videos; with interactive activities and quizzes, the course lasts two hours. Co-author and co-presenter, with equal screen time.

Regulators, insurers and public bodies — 2023 - 2026
Canada, United States, Switzerland

- Met insurance regulators in person in Canada, the United States and Switzerland to present the state of the research.
- Provided input, in person, to the Swiss Federal Statistical Office and the ETH Zurich AI Policy Lab on a governmental framework for fair predictive models in employment.
- Presented at the internal annual actuarial conferences of three insurers.
- Three companies contacted me for technical details on implementing the toolbox with their own data.

REFEREEING

European Actuarial Journal — 2026 – present

Annals of Actuarial Science — 2026 – present

Journal of Risk and Insurance — 2024 – present

SCHOLARSHIPS AND AWARDS

SEK equivalents at the European Central Bank reference rates of 14 October 2026 (1 CAD ≈ 7.01 SEK; 1 USD ≈ 9.97 SEK).

Award for Research — 2025, 2026
American Academy of Actuaries
Winner in 2026 (honorarium US$7 500, ≈ 75 000 SEK) for *A scalable toolbox for exposing indirect discrimination in insurance rates*; finalist in 2025 for *A fair price to pay: Exploiting causal graphs for fairness in insurance*. Annual international competition for papers by early-career scholars.

Travel Grant — 2026
NSERC General Fund
Travel grant to support conference participation and dissemination of research ($1 300 CAD, ≈ 9 100 SEK).

Actuarial Research Conference 2026 Travel Grant — 2026
American Academy of Actuaries
Travel grant to support conference participation and dissemination of research (US$1 500, ≈ 15 000 SEK).

Student travel grant — 2026
Big Data Research Center
Travel grant to support conference participation and dissemination of research ($2 000 CAD, ≈ 14 000 SEK).

Canada Graduate Research Scholarship – Doctoral — 2025
Natural Sciences and Engineering Research Council of Canada
National competition, $120 000 CAD (≈ 841 000 SEK) over three years ($40 000 per year, ≈ 280 000 SEK), held from May 2025.

Hickman Scholars Program — 2024, 2025, 2026
Society of Actuaries
International competition, US$20 000 (≈ 199 000 SEK) per year for three years (US$60 000 in total, ≈ 598 000 SEK). Every annual renewal is competitive.

Doctoral Training Scholarships — 2024
Fonds de recherche du Québec – Nature et technologies
Provincial competition, $100 000 CAD (≈ 701 000 SEK) offered over four years; held from May 2024 to May 2025, when the NSERC doctoral scholarship began.

Mitacs Accelerate — 2024
Mitacs
Research partnership with a large Canadian P&C insurer, $84 000 CAD (≈ 589 000 SEK) over three years (2024–2027), in which I am the lead student researcher. Selected after a scientific peer review and an internal Mitacs review.

Doctorate attraction grant Florent Toureille — 2023
Chaire d'actuariat de l'Université Laval
This competitive scholarship based on leadership and excellence amounts to $30 000 CAD (≈ 210 000 SEK) on three years.

Leadership and Engagement Scholarship — 2023
Université Laval
This competitive sports leadership scholarship amounts to $30 000 CAD (≈ 210 000 SEK) on three years.

Charles Brindamour Graduate Scholarship for Excellence — 2023
Université Laval, funded by Charles Brindamour (CEO of Intact Financial Corporation)
Institutional competition based on excellence, one award a year, $10 000 CAD (≈ 70 000 SEK).

François-Laviolette Scholarship — 2022
Université Laval
Institutional competition aimed at the development of responsible artificial intelligence, $7 500 CAD (≈ 53 000 SEK).

Grant of attraction to the Master's — 2021
Chaire d'actuariat de l'Université Laval
Grant recognizing the research potential for future Master's thesis students, amount of $15 000 CAD (≈ 105 000 SEK).

CAS Trust Scholarship — 2021
Casualty Actuarial Society
International competition, US$2 500 (≈ 25 000 SEK).

Alexander Graham Bell Canada Graduate Scholarship – Master's — 2021
Natural Sciences and Engineering Research Council of Canada
National competition, $17 500 CAD (≈ 123 000 SEK).

Lieutenant Governor's Youth Medal — 2018
Cégep de Lévis
Awarded to a student whose involvement and academic results stand out; nominated by the Cégep's teaching staff.

ACADEMIC COMMUNICATIONS

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice — 2026
Casualty Actuarial Society webinar
Sole presenter for a 90-minute webinar to 950 registrants **(Audience: end-users).**

The Trilemma of Fairness in P&C Pricing — 2026
61st Actuarial Research Conference (AAA invited session)
Invited presentation **(Audience: decision-makers).**

Profiling Actuarial Discrimination via Causal Decomposition — 2026
29th International Congress on Insurance: Mathematics and Economics, Sungkyunkwan University, Seoul (South Korea)
Academic presentation **(Audience: academics).**

Le trilemme de l'équité en tarification IARD — 2026
Association des actuaires en IARD du Québec (AAIARD)
Presentation to P&C actuaries **(Audience: end-users).**

Profiling Actuarial Discrimination via Causal Decomposition — 2026
STATQAM seminar, Université du Québec à Montréal
Invited seminar covering causal decomposition and confounding bias **(Audience: academics).**

Profiling Actuarial Discrimination via Causal Decomposition — 2026
Centre de recherche en données massives, Université Laval
Research talk **(Audience: academics).**

Profiling Actuarial Discrimination via causal decomposition — 2026
Actuarial and Financial Mathematics Conference, Brussels (Belgium)
Poster presenter **(Audience: academics).**

Bias in Assessing Financial Risk: Insights from the 2025 Research Award — 2026
American Academy of Actuaries
Invited panelist as finalist for the 2025 Award for Research on Bias in Assessing Financial Risk. Webinar to 445 actuaries and practitioners. **(Audience: end-users)**

A Causal Perspective on Fairness in Insurance Pricing — 2026
Department of Mathematics, Stockholm University
Invited seminar talk during research stay **(Audience: academics)**.

A Scalable Toolbox for Exposing Indirect Discrimination In Insurance Rates — 2026
ASTIN Bulletin Conference, Zurich (Switzerland)
Speaker at one of the fairness sessions of the first ASTIN Bulletin Conference **(Audience: academics).**

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice — 2025
Department of Statistics, University of Toronto
Invited seminar during research visit. **(Audience: academics)**

Operationalizing Fairness in Actuarial Pricing: From Principle to Practice — 2025
Casualty Actuarial Society Annual Meeting, Austin (United States) & Canada Connect Day, Toronto (Canada)
Invited presenter to over 150 (Austin) and over 100 (Toronto) actuaries and policymakers. **(Audience: end-users)**

Two invited presentations at the 60th Actuarial Research Conference — 2025
American Academy of Actuaries panel & Casualty Actuarial Society session, Toronto (Canada)
Guest panelist and presenter (with Marie-Pier and Arthur) to audiences from actuarial organizations, academia, industry, and policymakers. **(Audience: decision-makers)**

A Scalable Toolbox for Exposing Indirect Discrimination in Insurance Rates — 2025
Perspectives on Actuarial Risks in Talks of Young Researchers (PARTY), Liverpool (U.K.)
Presenter to 60 doctoral students from 29 countries. **(Audience: academics)**

Une boîte à outils pour dévoiler la discrimination indirecte dans les tarifs d'assurance 2025
Chaire ACTIONS & Milliman R&D, Paris (France)
Guest co-presenter (with Arthur) to audiences from academia and industry (80 participants at the Chaire ACTIONS). **(Audience: general)**

Perspective causale sur les discriminations dans les modèles prédictifs en assurance 2024
Institut intelligence et données
Guest presenter to over 100 people from academia. **(Audience: academics)**

A Fair price to pay: exploiting causal graphs for fairness in insurance 2024
27th International Congress on Insurance: Mathematics and Economics
Academic presentation in Chicago, IL, United States **(Audience: academics)**

A Fair price to pay: exploiting causal graphs for fairness in insurance 2024
Workshop on fairness and discrimination in insurance
Guest presenter to over 60 people from academia, practitioners and regulators. **(Audience: general)**

Perspective causale sur la discrimination indirecte en assurance générale 2024
Autorité des marchés financiers
Presentation to decision-makers at Québec's financial authority, followed by interviews with ten regulators on their needs. **(Audience: decision-makers)**

Équité et discrimination en assurance générales 2023
Intact Financial Corporation
Presentation to CEO, CLO and CDO of Intact Financial Corporation. **(Audience: decision-makers)**

Causal inference and Fairness in Insurance Pricing 2023
Insurance Data Science Conference
Presenter for Insurance Data Science Conference in London (UK). **(Audience: Academics)**

PUBLIC OUTREACH

L'avenir à travers une boule de cristal mathématique : Le travail d'un actuaire 2023-2026
Cégep de Lévis
Guest presenter for CEGEP students. **(Audience: Students)**

Pour une intelligence artificielle équitable en assurance 2024
Université Laval News
I collaborated for a scientific communication article. **(Audience: General Public)**

Perspective causale sur la discrimination indirecte en assurance générale 2024
Institut Intelligence et données
4th research competition on data intelligence. ≈ 50 graduate students **(Audience: general)**

Perspective causale sur la discrimination indirecte en assurance générale 2024
Institut de valorisation des données
I presented my research during IVADO Digital Future Day 2024. Over 250 people. **(Audience: general)**

L'IA pour calculer le prix de votre assurance auto: poétique ou pas éthique? 2023
Big Data Research Center
Presentation during the event *lightning talks 180* **(Audience: General)**
***Awards included:** Public's choice award

Artificial intelligence to prevent unfair discrimination in auto insurance 2023
IVADO
I authored an 800-word scientific communication article. **(Audience: General)**

ORGANIZING

Workshop on fairness and discrimination in insurance — 2026
Université Laval, 6–7 October 2026
Member of the organizing committee.

Quantact Student Day — 2025
Province-wide event, 3 October 2025
Organizer: recruited a five-member committee, coordinated seven presenters and hosted sixty students.

COMMITTEES

Institut Intelligence et Données, Université Laval — 2025 - present
Voting member of the board

Actuarial Science Graduate Program Committee of Université Laval — 2022 - present
Voting member of the Committee

Big Data Research Center of Université Laval — 2022 - 2025
Student voting member of the Committee

Actuarial Science Student Investment Fund of Université Laval — 2021 - 2024
Voting member of the committee

RESEARCH AFFILIATIONS

Institut Intelligence et Données; Big Data Research Center — 2021 - present
Université Laval

Quantact research laboratory; Centre de recherches mathématiques — 2021 - present
Montréal

OTHER

Roundtable on raising awareness about youth bullying — 2025
Mammouth hebdo - Téléquébec

Actuarial School representative at the open house — 2022 - 2023
Faculty of Science and Engineering

- Winter 2023 (≈ 30 visits from students accompanied by their parents)
- Fall 2022 (≈ 50 visits from students accompanied by their parents)